

Ordering fractions; adding mixed numbers

One common denominator settles both the order and the sum. [Cambridge insert]

LESSON 134–136

3 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 7.1–7.2 FRACTIONS

LESSON CLOCK

25'

30'

35'

25'

5

Comparing two fractions	25 min
Ordering a list	30 min
Adding mixed numbers	35 min
Subtracting with a regroup	25 min
Homework	5 min

LEARNING OBJECTIVES

- Compare two fractions by rewriting them over a common denominator.
- Order a list of fractions and place them on a number line.
- Add and subtract mixed numbers, converting the improper part back.
- Regroup one whole when the subtraction of the fraction parts will not go.

TEXTBOOK

National	pp. 389–394
Cambridge	Stage 7, pp. 68–75
Workbook	Exercise 7.1–7.2

01

Explanation

Comparing two fractions

Two fractions can be compared the moment they share a denominator: the one with the larger numerator is larger. Everything in this topic is that one sentence, applied twice.

PAIR	COMMON DENOMINATOR	REWRITTEN	LARGER
$\frac{3}{4}$ and $\frac{5}{7}$	28	$\frac{21}{28}$ and $\frac{20}{28}$	$\frac{3}{4}$
$\frac{5}{8}$ and $\frac{7}{12}$	24	$\frac{15}{24}$ and $\frac{14}{24}$	$\frac{5}{8}$
$\frac{4}{9}$ and $\frac{3}{7}$	63	$\frac{28}{63}$ and $\frac{27}{63}$	$\frac{4}{9}$
$\frac{2}{3}$ and $\frac{5}{8}$	24	$\frac{16}{24}$ and $\frac{15}{24}$	$\frac{2}{3}$

THE CROSS CHECK, FOR TWO FRACTIONS ONLY

For $\frac{a}{b}$ against $\frac{c}{d}$, compare ad with bc . For $\frac{3}{4}$ and $\frac{5}{7}$: $3 \cdot 7 = 21$ against $4 \cdot 5 = 20$, so the first is larger. It is the same arithmetic as the table, with the denominator left unwritten.

Ordering a list

For three or more fractions the cross check is no help — one denominator for the whole list is quicker. Order $\frac{2}{3}$, $\frac{3}{5}$, $\frac{7}{10}$, $\frac{5}{8}$.

FRACTION	OVER 120	AS A DECIMAL	PLACE
$\frac{3}{5}$	$\frac{72}{120}$	0.600	1st
$\frac{5}{8}$	$\frac{75}{120}$	0.625	2nd
$\frac{2}{3}$	$\frac{80}{120}$	0.667	3rd
$\frac{7}{10}$	$\frac{84}{120}$	0.700	4th

A BIGGER DENOMINATOR DOES NOT MEAN A BIGGER FRACTION

$\frac{7}{10}$ beats $\frac{5}{8}$ and $\frac{3}{5}$ beats nothing at all: the denominator alone decides nothing. Rewrite, then compare.

With directed numbers the order reverses on the negative side: $\frac{2}{3}$ is larger than $\frac{3}{5}$, so $-\frac{2}{3}$ is **smaller** than $-\frac{3}{5}$.

Adding mixed numbers

Add the whole parts, add the fraction parts over a common denominator, then carry any whole that the fraction part has produced.

$$2\frac{3}{4} + 1\frac{5}{6} = 3 + \frac{9}{12} + \frac{10}{12} = 3\frac{19}{12} = 4\frac{7}{12}$$

SUM	WHOLES	FRACTIONS	ANSWER
$3\frac{2}{5} + 2\frac{1}{3}$	5	$\frac{6}{15} + \frac{5}{15} = \frac{11}{15}$	$5\frac{11}{15}$
$1\frac{1}{2} + 2\frac{2}{3}$	3	$\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$	$4\frac{1}{6}$
$3\frac{5}{6} + 1\frac{3}{4}$	4	$\frac{10}{12} + \frac{9}{12} = \frac{19}{12}$	$5\frac{7}{12}$
$2\frac{7}{10} + 3\frac{4}{5}$	5	$\frac{7}{10} + \frac{8}{10} = \frac{15}{10}$	$6\frac{1}{2}$

NEVER LEAVE AN IMPROPER FRACTION IN THE ANSWER

$3\frac{19}{12}$ is a correct value but not a finished answer: $\frac{19}{12} = 1\frac{7}{12}$, so the whole part becomes 4.

Subtracting with a regroup

Subtraction is the same, until the fraction parts will not go. Then one whole is exchanged for a fraction of the same denominator — exactly the borrowing of column subtraction.

$4\frac{1}{3} - 1\frac{5}{6} = 4\frac{2}{6} = 3\frac{8}{6}, \text{ so } 3\frac{8}{6} - 1\frac{5}{6} = 2\frac{3}{6} = 2\frac{1}{2}$

DIFFERENCE	REGROUP	ANSWER
$5\frac{1}{4} - 2\frac{3}{4}$	$4\frac{5}{4} - 2\frac{3}{4}$	$2\frac{1}{2}$
$6 - 2\frac{3}{8}$	$5\frac{8}{8} - 2\frac{3}{8}$	$3\frac{5}{8}$
$7\frac{2}{5} - 3\frac{4}{5}$	$6\frac{7}{5} - 3\frac{4}{5}$	$3\frac{3}{5}$
$5\frac{1}{6} - 2\frac{3}{4}$	$4\frac{14}{12} - 2\frac{9}{12}$	$2\frac{5}{12}$

TAKE THE WHOLE ACROSS IN THE RIGHT PIECES

One whole is $\frac{4}{4}$ when the denominator is 4 and $\frac{12}{12}$ when it is 12. Change the denominators first, regroup second, or the exchange will be the wrong size.

02 Worked examples

EXAMPLE 1 Which is larger, $\frac{4}{9}$ or $\frac{3}{7}$?

① Common denominator 63.

② $\frac{4}{9} = \frac{28}{63}$ and $\frac{3}{7} = \frac{27}{63}$.

③ $28 > 27$.

Cross check: $4 \cdot 7 = 28 > 27 = 9 \cdot 3$ ✓

ANSWER

$\frac{4}{9}$ is larger

EXAMPLE 2 Work out $2\frac{3}{4} + 1\frac{5}{6}$.

① Wholes: $2 + 1 = 3$.

② Fractions over 12: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

③ $\frac{19}{12} = 1\frac{7}{12}$, so the answer is $4\frac{7}{12}$.

Estimate: $2.75 + 1.83 \approx 4.6$ ✓

ANSWER

$$4\frac{7}{12}$$

EXAMPLE 3 Work out $4\frac{1}{3} - 1\frac{5}{6}$.

① Over 6: $4\frac{2}{6} - 1\frac{5}{6}$.

② $\frac{2}{6} < \frac{5}{6}$, so regroup: $3\frac{8}{6}$.

③ $3\frac{8}{6} - 1\frac{5}{6} = 2\frac{3}{6} = 2\frac{1}{2}$.

ANSWER

$$2\frac{1}{2}$$

03 Interactive model

Run the model on the pair 9 and 7 last: the extra factors are the other denominator, and the class sees why unrelated denominators multiply.

Building the common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
4	$2 \cdot 2$	$\times 3$
6	$2 \cdot 3$	$\times 2$

LOWEST COMMON DENOMINATOR

$$2 \cdot 2 \cdot 3 = 12$$

The shared factor 2 is counted once, not twice.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mixed number	Aralash son	Смешанное число
Improper fraction	Noto'g'ri kasr	Неправильная дробь
Proper fraction	To'g'ri kasr	Правильная дробь
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
To order	Tartiblamq	Упорядочить
Ascending order	O'sish tartibida	По возрастанию
To regroup a whole	Butundan olmoq	Занимать из целой части

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To compare two fractions you first make equal the:

numerators

denominators

wholes

decimals

The larger of $\frac{5}{8}$ and $\frac{7}{12}$ is:

$\frac{5}{8}$

$\frac{7}{12}$

they are equal

cannot tell

$2\frac{3}{4} + 1\frac{5}{6}$ equals:

$$3\frac{19}{12}$$

$$4\frac{7}{12}$$

$$3\frac{8}{10}$$

$$4\frac{1}{2}$$

In $6 - 2\frac{3}{8}$ the whole 6 becomes:

$$5\frac{8}{8}$$

$$6\frac{8}{8}$$

$$5\frac{1}{8}$$

$$5$$

Of $-\frac{2}{3}$ and $-\frac{3}{5}$ the smaller is:

$$-\frac{3}{5}$$

$$-\frac{2}{3}$$

equal

neither

The lowest common denominator of 9 and 7 is:

16

63

9

126

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Which is larger, $\frac{3}{4}$ or $\frac{5}{7}$?

ANSWER $\frac{3}{4}$

2. Which is larger, $\frac{5}{8}$ or $\frac{7}{12}$?

ANSWER $\frac{5}{8}$

3. Which is larger, $\frac{2}{3}$ or $\frac{5}{8}$?

ANSWER $\frac{2}{3}$

4. $1\frac{1}{4} + 2\frac{1}{4}$

ANSWER $3\frac{1}{2}$

5. $3\frac{2}{5} + 1\frac{1}{5}$

ANSWER $4\frac{3}{5}$

6. $5\frac{3}{4} - 2\frac{1}{4}$

ANSWER $3\frac{1}{2}$

7. The lowest common denominator of 6 and 8

ANSWER 24

Medium · the standard question

7 PROBLEMS

1. $2\frac{3}{4} + 1\frac{5}{6}$

ANSWER $4\frac{7}{12}$

2. $3\frac{5}{6} + 1\frac{3}{4}$

ANSWER $5\frac{7}{12}$

3. $4\frac{1}{3} - 1\frac{5}{6}$

ANSWER $2\frac{1}{2}$

4. $6 - 2\frac{3}{8}$

ANSWER $3\frac{5}{8}$

5. $7\frac{2}{5} - 3\frac{4}{5}$

ANSWER $3\frac{3}{5}$

6. Order $\frac{3}{5}$, $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{8}$ upwards

ANSWER $\frac{3}{5}$, $\frac{5}{8}$, $\frac{2}{3}$, $\frac{7}{10}$

7. Which is larger, $-\frac{2}{3}$ or $-\frac{3}{5}$?

ANSWER $-\frac{3}{5}$

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{7}{10} + 3\frac{4}{5}$

ANSWER $6\frac{1}{2}$

2. $5\frac{1}{6} - 2\frac{3}{4}$

ANSWER $2\frac{5}{12}$

3. $1\frac{3}{8} + 2\frac{1}{6} + 1\frac{1}{4}$

ANSWER $4\frac{19}{24}$

4. $10 - 3\frac{2}{7} - 2\frac{5}{7}$

ANSWER 4

5. Find a fraction between $\frac{5}{9}$ and $\frac{4}{7}$

ANSWER $\frac{71}{126}$

6. A recipe needs $2\frac{1}{3}$ cups and $1\frac{3}{4}$ cups: the total

ANSWER $4\frac{1}{12}$ cups

7. Which is larger, $\frac{12}{17}$ or $\frac{7}{10}$?

ANSWER $\frac{12}{17}$, since $120 > 119$

07 Homework

Homework — 5 tasks

Every answer as a mixed number in lowest terms, with the common denominator shown.

1. Which is larger, $\frac{7}{9}$ or $\frac{5}{6}$? Show the rewriting.
2. Order $\frac{1}{2}$, $\frac{4}{9}$, $\frac{5}{12}$ and $\frac{7}{18}$ in ascending order.
3. Work out $3\frac{2}{3} + 2\frac{3}{4}$.
4. Work out $5\frac{1}{4} - 2\frac{2}{3}$.
5. A plank 4 m long has $1\frac{5}{8}$ m sawn off. How much is left?